Spotify: discovering and explaining groups

Recursive UMAP and HDBSCAN, explained through a computed demonstration

The people behind the use case

People	Ana and Luis play Izal and Vetusta Morla. Marta plays Charlotte de Witte.
Group	Artist-play counts could group Ana and Luis together, apart from Marta.
Why	An explanation could identify Spanish indie listening as the group's main signal.
Possible use	Suggest Love of Lesbian to Ana if Luis listens to them.

Illustrative artist example. The computed demo uses behaviour, not artist counts

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Technical details

A concrete listening example

Ana and Luis share artists. Can we discover their group and explain what makes it different?

Follow the same people

Each next step explains what the algorithm would mean for Ana, Luis and Marta.

Illustrative story. The calculated charts use synthetic behaviour features, without artist counts or named users.

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All listeners | Computed preset | Balanced · mcs 100 | Open explorer ↗

Computed demo · synthetic listeners

Source: Spotify Engineering · Educational reconstruction

← / → steps · colours identify local groups · no Spotify data or internal parameters

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Original features

$X \in \mathbb{R}^{684 \times 6}$

Unlabelled observations

Listener	Minutes	Sessions	Skips	Discovery	Playlists	Night
L001	222	11.4	10%	17%	52%	72%
L002	124	6.6	46%	64%	41%	29%
L003	32	1.8	28%	27%	75%	8%
L004	127	6.6	42%	66%	32%	36%

Numeric features, no prescribed segment labels

684 rows × 6 features

StandardScaler $z_{ij} = (x_{ij} - \mu_j) / \sigma_j$ Fit once on the full demo population

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Technical details

An original feature vector for every listener

Ana's listening becomes a numeric row. Here we compute six behaviour features. Artist counts would be extra inputs.

Listener L001

Minutes	222	Sessions	11.4
Skips	10%	Discovery	17%
Playlists	52%	Night	72%

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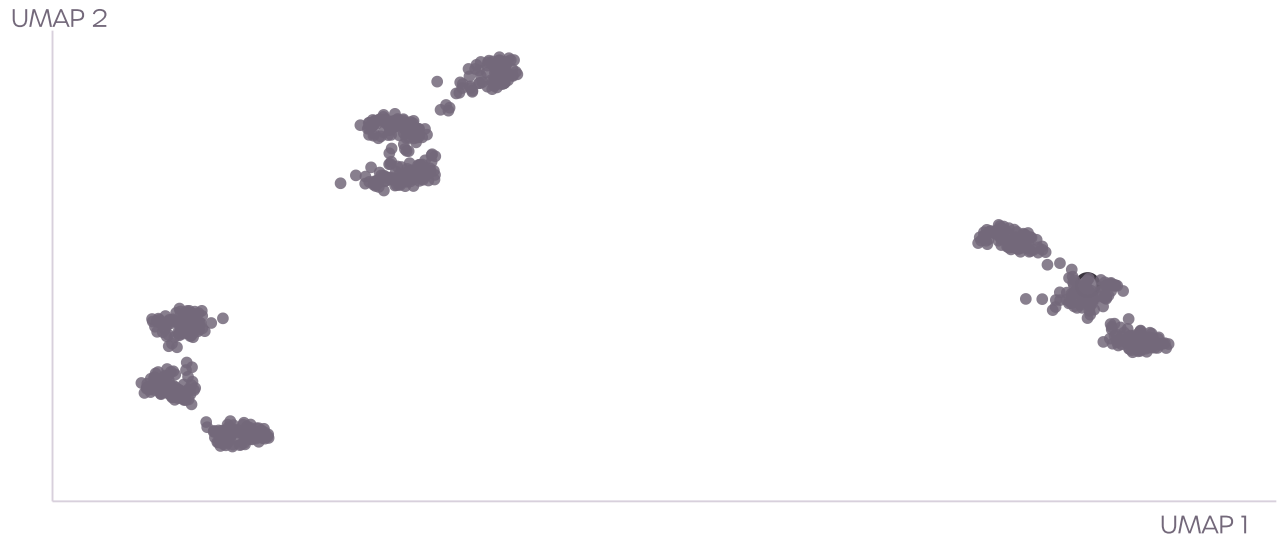
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Computed UMAP coordinates

3 clusters · 0 noise · 684 listeners



Display transition only. Point movement does not reproduce the optimizer's iterations.

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Technical details

UMAP makes neighbourhoods visible

With suitable artist features, Ana and Luis could appear nearby because they share artists. UMAP displays similarity, not a genre label.

684 × 6 → 684 × 2

10-neighbour trustworthiness: 0.975

Local-neighbourhood diagnostic, not segment validity.

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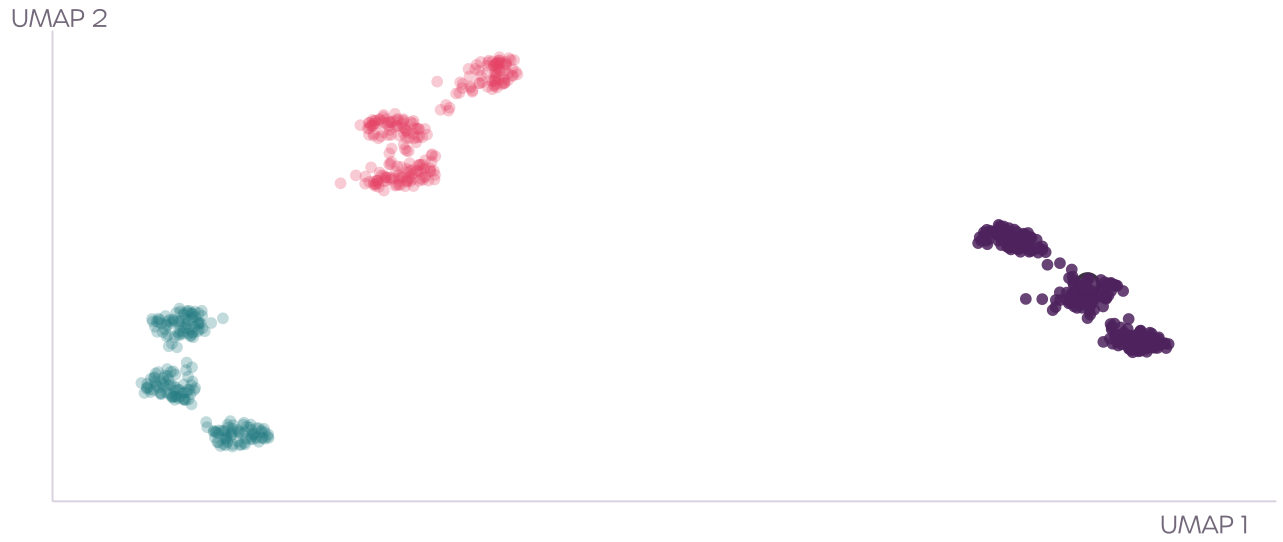
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Spotify

1 Example | 2 Features | 3 UMAP | 4 HDBSCAN | 5 Re-embed | 6 Hierarchy | 7 XGBoost | 8 SHAP | 9 Explainability

Computed UMAP coordinates

3 clusters · 0 noise · 684 listeners



C1 221 | C2 241 | C3 222 | Noise 0

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Technical details

HDBSCAN discovers density-based groups

A dense group could contain Ana and Luis, while Marta falls elsewhere. The colours show discovered groups. Our chart uses synthetic behaviour data.

Inspect group: C1 · 221 listeners

32.3% of this node

Noise: 0 / 684

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A new fit, not a magnified plot



New coordinates come from a new UMAP fit on the selected original rows.

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Technical details

Recursion starts from the original feature rows

Analyse Ana's group again using its original rows. This could reveal finer habits within her broader group.

Parent cluster: C1 · 221 rows

Inspect C1 child fit

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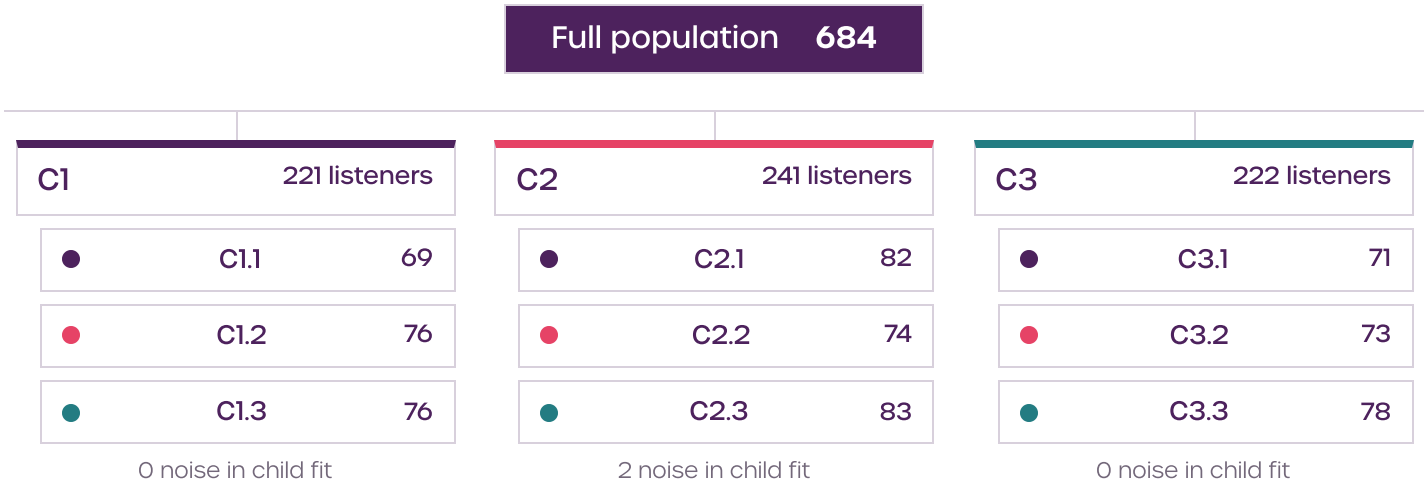
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Computed cluster hierarchy



Noise stays unassigned. Child counts can sum to less than their parent.

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Technical details

The hierarchy records each discovery step

Ana's broad group could split into subgroups. The tree records which users enter each new fit. Click a branch to inspect its computed population.

4 computed fits

Each child has its own UMAP coordinates and HDBSCAN assignments.

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One-versus-rest training

Original features X
684 rows × 6

Discovered labels
y = 1 for C1
y = 0 otherwise

Binary XGBoost
90 trees · depth 3
75% training rows

Held-out fidelity
F1 0.991
AUC 1.000

One model per cluster. Embedding coordinates are excluded.

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Technical details

A supervised model learns discovered membership

XGBoost learns to predict Ana's group versus everyone else, using the original features and discovered labels.

Membership model: C1 versus rest

513 training / 171 held-out

The split tests label reproduction, not new-user cluster discovery.

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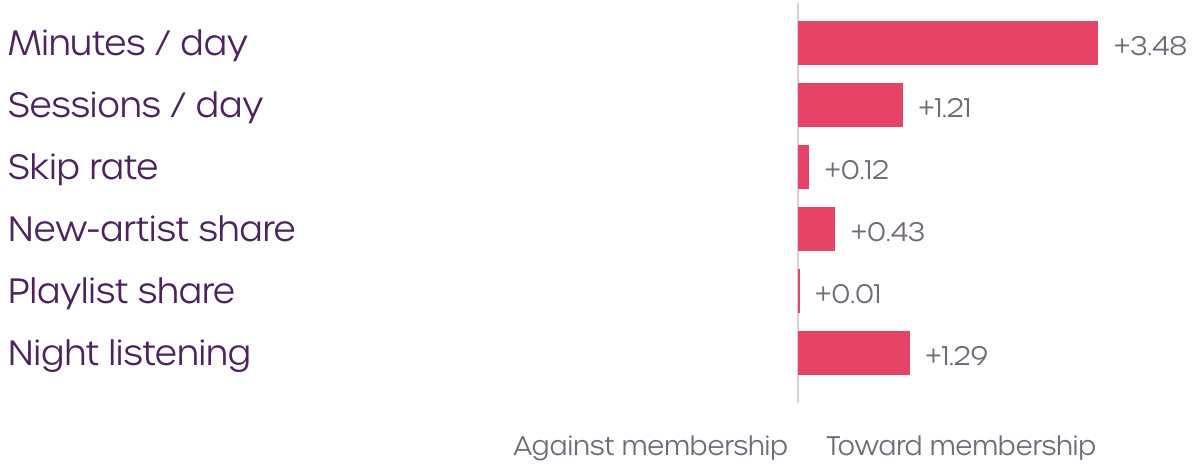
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Calculated TreeSHAP contributions

This listener | Mean |SHAP| | Signed contribution to membership log-odds



Base -0.80 + Σ SHAP 6.55 = log-odds 5.75

P(C1) = 99.7%

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Technical details

SHAP explains the membership predictor

SHAP explains a listener's group score. Indie listening could matter with artist inputs. These bars explain behaviour.

Explain membership in: C1

Listener: L001 · held-out

Held-out listener. Independent binary scores need not sum to 1.

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Why interpretation is needed

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Technical details

What the results mean

Group C1 Listener L001

Across listeners

Minutes / day and Night listening have the largest average effects.

This listener

Minutes / day = 222 is the strongest push toward C1.

Model explanations do not prove causes or validate segments.

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